

# Adaptive Study Planning for Self-Directed Learners

*Presentation Script*



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|--------------------|----------------------------------------------------|
| <b>Candidate</b>   | Rohit Saji                                         |
| <b>Programme</b>   | M.Tech Data Science & AI                           |
| <b>Institution</b> | PES University (Great Learning)                    |
| <b>Event</b>       | First Guidance Call · 31 May 2026                  |
| <b>Format</b>      | 6 slides · ~9–10 min speaking + 3–5 min discussion |

## HOW TO USE THIS SCRIPT

Read through this two or three times before presenting — you don't need to memorize it word for word, just get comfortable with the flow. Each slide shows what's on the slide, what to say (simple, conversational words), and a time target. Total: about 9–10 minutes speaking plus 3–5 minutes of discussion. Remember the guide is a PM / SE type, not an algorithm researcher, and he's hearing 13 projects back-to-back — so keep it crisp, structured, and system-focused.

## WHAT'S INSIDE

- Slides 1–6 — on-slide cues plus the word-for-word script and a time target each
- Closing line
- Faculty Q&A — seven prepared answers (know them; don't read them)
- Appendix — the 4-stage synthetic data pipeline, for your reference during Q&A

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SLIDE 1

≈ 90 sec

## Title, Problem & Solution

### ON THE SLIDE

- Title: Adaptive Study Planning for Self-Directed Learners
- Your name and programme
- Problem (2 lines)
- Proposed solution (2 lines)

### WHAT YOU SAY

Good morning. My name is Rohit Saji, M.Tech Data Science, fourth semester.

My project is about making study plans smarter for people who study on their own.

Here's the problem. When someone is studying for an exam or an online course, they make a plan — "I'll study 2 hours a day, finish by this date." But within a few days, they're behind. Or sometimes ahead. The plan doesn't match reality anymore, and they have no way to know how far off they are.

What I'm proposing is a study planning system that watches how you actually study, learns your real pace, detects if your habits change, and adjusts your schedule automatically. So instead of a static plan that breaks on day three, you get a plan that keeps up with you.

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SLIDE 2

≈ 90 sec

## Literature Survey Overview

### ON THE SLIDE

- "13 papers · 6 themes · since 2024" stat callout
- Table with 6 themes, paper counts, and one key finding each
- Gap statement at the bottom

### WHAT YOU SAY

Since the zeroth review, I've done a deeper literature survey. I've gone through 13 papers from 2024 to 2026, covering six research areas relevant to this project.

Let me walk through what I found.

For learner modeling, hierarchical Bayesian methods work well even with very small datasets — one study validated this with just 312 participants. This is important because my system needs to learn from one person's data, not thousands.

For detecting behavioral changes — like when someone suddenly slows down or stops studying — there's strong evidence from learning analytics that 88 percent of students who disengage show detectable warning signals before it happens. On the methods side, CUSUM and EWMA from statistical process control are well-established for this type of detection.

For prediction, Gaussian Process regression achieved 0.89 correlation on educational data, and it gives you confidence intervals — not just a single number, but a range.

For schedule generation, constraint-based and dynamic programming approaches both show promise for building adaptive paths.

The gap I identified is clear: all these techniques exist separately. Nobody has connected them into one closed-loop system that watches, learns, predicts, and adjusts for a self-directed learner.

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SLIDE 3

## Five Research Areas with Candidates

≈ 90 sec

### ON THE SLIDE

- 5 research areas listed vertically
- 4 marked as resolved (with candidate algorithms)
- 1 marked as open (evaluation)
- “was open” tags on areas 2 and 4

### WHAT YOU SAY

At the zeroth review, I had identified five research questions for this project and had candidates for only two of them. Since then, I've resolved four of the five.

For learning study pace, I'm comparing hierarchical Bayesian calibration against a simple moving average baseline.

For detecting behavioral changes — which was an open question at zeroth — my reading points to CUSUM, EWMA, and critical slowing down indicators as the candidates to compare.

For predicting when someone will finish, I'm comparing Gaussian Process regression against linear extrapolation — the GP gives uncertainty bands, which is the key advantage.

For automatically adjusting the schedule — also open at zeroth — I'm looking at constraint-based generation with role-based material phasing versus dynamic programming versus rule-based approaches.

The fifth area — evaluation methodology — is the one I'd like your input on. I'm thinking of a before-and-after comparison: use the static planner from Phase 1 for a period, then switch to the adaptive planner in Phase 2, and compare schedule adherence and prediction accuracy on the same user. But I'm not sure if this design is rigorous enough. What would you suggest?

*[Pause briefly for the guide to react. If he responds now, engage. If he nods or says “we can discuss later,” continue.]*

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SLIDE 4

## Proposed System Architecture

≈ 90 sec

### ON THE SLIDE

- Block diagram with two zones
- Phase 1: solid boxes (Session Logger, Event Store, Static Generator, Dashboard)
- Phase 2: dashed boxes (“could be” Bayesian, CUSUM, GP, constraint-based)
- Data flow arrows

### WHAT YOU SAY

Here's the system architecture I'm proposing. I'd like your input on this.

In Phase 1, I build the platform — four components. A session logger that records every study session. An event store that persists the data locally. A static schedule generator that creates a study plan from

the user's inputs — their deadline, materials, and available hours. And a basic dashboard to visualize progress.

In Phase 1, the data flows in one direction — from logging through to the dashboard. No feedback, no intelligence. It's a static planner.

In Phase 2, I add the intelligent layer — shown here as dashed boxes because these are the components I'm still researching. A pace calibration engine — could be Bayesian. A behavior monitor — could be CUSUM. A progress predictor — could be GP regression. And an adaptive regenerator that takes the calibrated pace and adjusts the schedule.

The key difference is that Phase 2 closes the loop. The system watches how you study, updates its model, and regenerates your schedule. Phase 1 is the baseline, Phase 2 is the intelligence.

Does this architecture make sense to you? Is there anything you'd change?

*[Pause for feedback. This is where the PM/SE guide is most likely to engage.]*

**5****SLIDE 5**

## Comparison Methodology

≈ 90 sec

### ON THE SLIDE

- 4-stage pipeline as horizontal flow with timeline
- Each stage: title + short description
- Learner profiles listed under Stage 1
- “Ground truth” callout

### WHAT YOU SAY

For comparing the candidate algorithms, I've planned a four-stage methodology.

In the first two weeks, I define learner profiles from the literature — what does a consistent learner look like statistically? What about someone who starts strong and slows down? I'll create four profiles: consistent, declining, improving, and irregular.

In weeks three and four, I build a data generator that creates synthetic study sessions based on these profiles. The key advantage here is that I control the ground truth. I know exactly where a behavioral shift happens because I planted it. So when my algorithms try to detect that shift, I can measure exactly how fast and how accurately they find it.

In weeks five and six, I run all the candidate algorithms on the same generated data and compare them using standard metrics — prediction error for pace learning, detection latency for behavior change, confidence interval calibration for prediction.

And in weeks seven and eight, once the app is built and I've been logging real study sessions, I validate the findings against real data to make sure the synthetic results hold up.

The data generator itself becomes a research contribution — it's a configurable study behavior simulator grounded in literature distributions.

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SLIDE 6

## Phase 1 Roadmap across 3 Reviews

≈ 60 sec

### ON THE SLIDE

- Horizontal timeline: 6 events from May 31 to Jul 5–6
- Review boxes with code percentages (30%, 80%, 100%)
- Deliverables listed under each review
- “Does this plan align?” callout at bottom

### WHAT YOU SAY

Here's how I plan to deliver Phase 1 across the three reviews.

At Review 1, which should be around June 7th, I'll present the formal abstract, the architecture diagram, all five research areas with algorithm candidates, expected outcomes, and references. On the code side, I'll have the session logging and event store working — about 30 percent of the system.

At Review 2, around June 21st, I'll show the detailed algorithm design, intermediate comparison results from the synthetic data pipeline, and about 80 percent of the code — the static scheduler, dashboard, and the comparison scripts.

At Review 3, the final review around July 5th, everything comes together — full experimental results, model comparison across all four learner profiles, a working demo of the static planner, and a draft journal paper. That's 100 percent of the Phase 1 code.

Does this plan align with your expectations for Phase 1? Is there anything you'd want me to focus on differently?

## Closing

≈ 15 sec

Thank you for your time. I'm looking forward to your guidance as I work through this. Please let me know if you'd like me to adjust any part of this plan.

## Faculty Q&A — prepared answers

*Keep these in mind. Don't read them — just know the key points.*

### Q. “How will one person's data be enough?”

A. The system is designed to personalize to each individual user — it builds a model for one person based on their own sessions. My own data validates that the system works end-to-end on a real user. For algorithm comparison, I use synthetic data with configurable learner profiles to get enough volume for meaningful comparison — that's the four-stage methodology I showed.

### Q. “What's your base paper?”

A. Two papers are central. Zanellati's survey on hybrid knowledge tracing models from IEEE, which validates the Bayesian approach for learner modeling. And Perez-Suay's work applying Gaussian Processes to student data, which achieves 0.89 correlation. I'm building on both directions — Bayesian for pace calibration, GP for progress prediction.

**Q. “Why not use an existing dataset?”**

A. Existing educational datasets like OULAD have click-stream data — page views, forum posts — but not session-level pace data in the format my algorithms need. I need planned-versus-actual duration per session, which no existing dataset provides. The synthetic data generator lets me create exactly this format with controlled ground truth, and then I validate against real data from the app.

**Q. “What makes this different from a todo app or calendar?”**

A. A calendar doesn't know how fast you study. If you planned 2 hours for a chapter but you consistently take 3 hours, a calendar won't tell you that your deadline is unrealistic. My system watches your actual pace, detects when it changes, and adjusts the remaining schedule automatically.

**Q. “What are the specific evaluation metrics?”**

A. That's the open research question I'd like your input on. I'm considering schedule adherence — how often the learner finishes sessions within the planned time. Prediction accuracy — how close the predicted finish date is to reality. And pace ratio drift — how much the estimated pace diverges from actual pace over time.

**Q. “How is Phase 2 different from Phase 1?”**

A. Phase 1 builds a static planner — it generates a schedule from your inputs but never updates it. The algorithms are tested offline, not in the app. Phase 2 plugs the best algorithms into the app, closing the feedback loop. The system now watches, learns, and adjusts automatically. The comparison is: does the adaptive version from Phase 2 help the learner stick to their plan better than the static version from Phase 1?

**Q. If asked about timeline concerns (too ambitious / too compressed):**

A. The four-stage methodology is designed to fill the full Phase 1 duration. Each stage produces a visible deliverable for the next review. And the early stages — profile definition and generator building — don't depend on the app being complete, so research and development can run in parallel.

## Appendix — 4-Stage Synthetic Data Pipeline

*For your reference only — not for presenting. Use it to add depth during Q&A.*

### Stage 1 (Weeks 1–2) — Define learner profiles from literature.

Read papers and extract realistic numbers — session duration distributions (lognormal, 30–90 minutes), sessions per week (3–7), pace variability by material type, and behavioral shift characteristics (onset, magnitude, duration). Define 4 profiles: Consistent (steady pace, no shifts), Declining (starts strong, slows mid-course), Improving (slow start, speeds up), and Irregular (unpredictable, high variance).

### Stage 2 (Weeks 3–4) — Build the data generator.

Code that takes a learner profile and generates a dataset of 100–500 study sessions. Each session records: material studied, planned duration, actual duration, time of day, day of week, and session index. Key property: ground truth is embedded — you KNOW where regime shifts are because you planted them, and you KNOW the real pace because you defined it, so algorithm accuracy is precisely measurable. Parameters are configurable for sensitivity analysis.

### Stage 3 (Weeks 5–6) — Run comparisons.

Run all candidate algorithms on the same generated datasets: Area 1 — Hierarchical Bayesian vs Simple Moving Average (prediction error on held-out sessions); Area 2 — CUSUM vs EWMA vs CSD (detection latency vs false-alarm rate); Area 3 — GP regression with ARD vs Linear extrapolation (confidence-interval calibration); Area 4 — Constraint-based vs DP vs Rule-based (schedule adherence after adjustment). Run each across all 4 profiles and show which algorithm wins per condition and whether any dominates.

### Stage 4 (Weeks 7–8) — Validate with real data.

Run the winning algorithms on actual study sessions from the app. Check: do the synthetic results hold up? Are there real-world patterns the profiles missed? Do the GP confidence intervals contain the actual outcomes? This bridges offline research to the working system.